

Normal	TimeIT	TGC	HD	MTG	RTL	Egoschema Full	VideoMME w/o subs	Charades-STA R@1 IOU=0.5	QVHighlights Hit@1
✓						56.6	42.6	8.0	24.4
✓	✓					57.8	43.6	32.2	25.2
✓	✓	✓				58.3	44.0	39.1	33.9
✓	✓	✓	✓			59.8	44.9	41.9	43.8
✓	✓	✓	✓	✓		60.0	45.1	45.8	48.3
✓	✓	✓	✓	✓	✓	60.0	46.3	48.7	54.1

Table 5: Performance results of the ablation study on different components of TimePro. We use 82K normal training data as the baseline. TimeIT refers to the training data with five task types from Ren et al. (2024), TGC refers to Temporal Grounded Caption, HD refers to Highlight Detection, MTG refers to Multi-format Temporal Grounding, and RTL refers to Reasoning Temporal Localization.

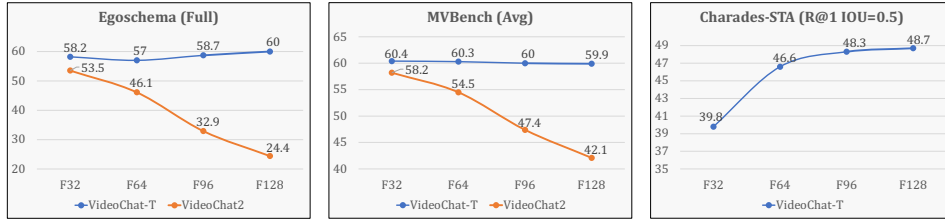


Figure 5: Performance of VideoChat-T with varying input frame numbers. As the number of input frames increases, the performance of VideoChat-T shows an upward trend in both long video QA and temporal grounding tasks. Due to the over low temporal grounding performance of VideoChat2, its curve is omitted.

gressively added subsets of TimePro, not only did the model’s performance on temporal grounding tasks show a stable and significant improvement, but we also observed a noticeable upward trend in performance on long video benchmarks. This to some extent corroborates that temporal grounding centric tasks have a positive impact on long video understanding.

Impact of frames. To investigate the impact of input frame count on model performance, we conducted an ablation study. Figure 5 illustrates the scalability of our model’s performance with respect to input frame count. VideoChat-T demonstrates good stability as the input frame count varies, and its performance in long video QA and temporal grounding tasks improves with an increase in frame count. In contrast, the baseline model, VideoChat2, exhibited catastrophic performance degradation when the frame count was significantly increased. As the input frame count increases, the number of visual encoding tokens grows linearly. Excessive visual token input imposes an additional computational burden on the temporal modeling of the LLM. TimeSuite mitigates this by employing Token Shuffle to reduce the number of tokens, ensuring the stable operation of the model.

5 CONCLUSION

In this paper, we have introduced TimeSuite, a collection of new designs from perspectives of efficient architecture, high-quality data, and new instruction tuning task, to achieve long video understanding by fine-tuning short video MLLMs with temporal grounding-centric data. We address the computational challenges of processing long videos by introducing token shuffle to compress visual tokens. We also propose the TAPE for adaptive position encoding, enhancing the temporal awareness of visual representation. Additionally, our designed Temporal Grounded Caption training task ensure MLLMs to build correspondence between grounded segments and detailed caption, while the TimePro dataset provide comprehensive instruction tuning data for learning more effective temporal perception capability. Experimental results demonstrate that VideoChat-T significantly improves long video understanding, with notable performance gains on Egoschema and VideoMME. Furthermore, VideoChat-T exhibits strong zero-shot temporal grounding capabilities, significantly outperforming the previous MLLMs on temporal grounding. Overall, our TimeSuite provides effective designs for short MLLMs to enhance their performance on temporal grounding and long video QA. We hope our TimeSuite could yield some insights on designing long video MLLMs.

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